

Roll Number:2403A51457
AI-Assisted-Coding(6.4)Assignment

TASK-1:

The screenshot shows a Visual Studio Code editor with a Python file named `task-1.py`. The code defines a `Student` class with attributes `name`, `roll_number`, and `marks`. It includes methods `__init__`, `display_details`, and `is_passed` (which checks if marks are greater than or equal to 40). The main part of the code takes user input for name, roll number, and marks, then prints the student details and their status.

```
1 class Student:
2     def __init__(self, name, roll_number, marks):
3         self.name = name
4         self.roll_number = roll_number
5         self.marks = marks
6
7     def display_details(self):
8         print(f"Name: {self.name}")
9         print(f"Roll Number: {self.roll_number}")
10        print(f"Marks: {self.marks}")
11
12    def is_passed(self, average=40):
13        if self.marks >= average:
14            print("Status: Passed")
15            return True
16        else:
17            print("Status: Failed")
18            return False
19
20 # Taking input from user
21 name = input("Enter student name: ")
22 roll_number = input("Enter roll number: ")
23 marks = float(input("Enter marks: "))
```

The terminal output shows the execution of the code with the following input and output:

```
PS C:\Users\DELL\6.4_ai_assisted> & "C:\Program Files\Python313\python.exe" c:\Users\DELL\6.4_ai_as
sisted\task-1.py
Enter student name: shiva
Enter roll number: 1457
Enter marks: 76

Student Details:
Name: shiva
Roll Number: 1457
Marks: 76.0
Result:
Status: Passed
PS C:\Users\DELL\6.4_ai_assisted>
```

TASK-2:

Generate a python code for the first two lines of a for loop to iterate through a list of numbers, to calculate and print the square of even numbers only, A complete loop generated with conditional logic (if number % 2 == 0). even number between 0-20

The screenshot shows a VS Code editor with a sidebar on the left containing a list of tasks: task-1.py, task-2, task-3, task-4, and task-5. The main editor area displays the code for task-2.py, which is a Python script that generates squares of even numbers from 0 to 20. The terminal at the bottom shows the execution output, listing the squares of 0, 2, 4, 6, 8, 10, 12, 14, 16, 18, and 20. The terminal prompt is PS C:\Users\DELL\6.4_ai_assisted>.

```
task-2 > ...
1 numbers = list(range(0, 21))
2 for number in numbers:
3     if number % 2 == 0:
4         print(f"Square of {number} is {number ** 2}")

sisted/task-2
Square of 0 is 0
Square of 2 is 4
Square of 4 is 16
Square of 6 is 36
Square of 8 is 64
Square of 10 is 100
Square of 12 is 144
Square of 14 is 196
Square of 16 is 256
Square of 18 is 324
Square of 20 is 400
PS C:\Users\DELL\6.4_ai_assisted>
```

TASK-3:

The screenshot shows a VS Code editor with task-3.py open. The code defines a BankAccount class with methods for deposit, withdrawal, and displaying the balance. The terminal shows the execution of the program, which prompts the user for account holder name, initial balance, and then performs a deposit and a withdrawal. An AI chat window on the right provides a summary of the code's functionality. The terminal prompt is PS C:\Users\DELL\6.4_ai_assisted>.

```
task-3 > ...
1 class BankAccount:
2     def __init__(self, account_holder, balance=0):
3         self.account_holder = account_holder
4         self.balance = balance
5
6     def deposit(self, amount):
7         if amount > 0:
8             self.balance += amount
9             print(f"Deposited: {amount}. New balance: {self.balance}")
10        else:
11            print("Deposit amount must be positive.")
12
13    def withdraw(self, amount):
14        if amount > self.balance:
15            print("Insufficient balance. Withdrawal denied.")
16        elif amount <= 0:
17            print("Withdrawal amount must be positive.")
18        else:
19            self.balance -= amount
20            print(f"Withdrew: {amount}. New balance: {self.balance}")
21
22    def display_balance(self):
23        print(f"Account Holder: {self.account_holder}")

PS C:\Users\DELL\6.4_ai_assisted> & "C:\Program Files\python313\python.exe" c:/Users/DELL/6.4_ai_assisted/task-3
Enter account holder name: Shiva
Enter initial balance: 50000
Account Holder: Shiva
Current Balance: 50000.0
Enter amount to deposit: 20000
Deposited: 20000.0. New balance: 70000.0
Enter amount to withdraw: 36000
Withdrawn: 36000.0. New balance: 34000.0
Account Holder: Shiva
Current Balance: 34000.0
PS C:\Users\DELL\6.4_ai_assisted>
```

Using python Create a class called BankAccount with attributes account_holder and balance.to deposit(), withdraw(), and check for insufficient balance.Functional class with complete method definitions using if conditions and self attributes. Code should prevent overdrawn.And also we should enter the deposit,withdraw amount

```
class BankAccount:
    def __init__(self, account_holder, balance=0):
        self.account_holder = account_holder
        self.balance = balance

    def deposit(self, amount):
        if amount > 0:
            self.balance += amount
            print(f"Deposited: {amount}. New balance: {self.balance}")
        else:
            print("Deposit amount must be positive.")

    def withdraw(self, amount):
        if amount > self.balance:
            print("Insufficient balance. Withdrawal denied.")
        elif amount <= 0:
            print("Withdrawal amount must be positive.")
        else:
            self.balance -= amount
            print(f"Withdrew: {amount}. New balance: {self.balance}")
```

TASK-4:

```
task-4.py
1 students = []
2 n = int(input("Enter number of students: "))
3
4 for _ in range(n):
5     name = input("Enter student name: ")
6     score = float(input("Enter student score: "))
7     students.append({"name": name, "score": score})
8
9 i = 0
10 print("\nStudents who scored more than 75:")
11 while i < len(students):
12     if students[i]["score"] > 75:
13         print(students[i]["name"])
14     i += 1
```

```
PS C:\Users\DELL6.4_ai_assisted> "C:\Program Files\Python313\python.exe" c:\Users\DELL6.4_ai_assisted/task-4
Enter number of students: 3
Enter student name: shiva
Enter student score: 76
Enter student name: siddu
Enter student score: 64
Enter student name: sateesh
Enter student score: 71

Students who scored more than 75:
shiva
```

TASK-5:

```
task-5.py
1 class ShoppingCart:
2     def __init__(self):
3         self.items = [] # Each item is a dict: {"name": str, "price": float,
4
5     def add_item(self, name, price, quantity=1):
6         self.items.append({"name": name, "price": price, "quantity": quantity
7         print(f"Added {quantity} x {name} at {price} each.")
8
9     def remove_item(self, name):
10        removed = False
11        for item in self.items:
12            if item["name"] == name:
13                self.items.remove(item)
14                print(f"Removed {name} from cart.")
15                removed = True
16                break
17        if not removed:
18            print(f"{name} not found in cart.")
19
20    def calculate_total(self):
21        total = 0
22        for item in self.items:
23            total += item["price"] * item["quantity"]
```

```
PS C:\Users\DELL6.4_ai_assisted> "C:\Program Files\Python313\python.exe" c:\Users\DELL6.4_ai_assisted/task-5
Enter number of items to add: 4
Enter item name: Eggs
Enter item price: 130
Enter item quantity: 1
Added 1 x Eggs at 130.0 each.
Enter item name: Bread
Enter item price: 49
Enter item quantity: 2
Added 2 x Bread at 49.0 each.
Enter item name: Peanut Butter
Enter item price: 149
```

Output:

```
6.4_AI_ASSISTED
task-1.py
task-2
task-3
task-4
task-5

PS C:\Users\DELL\6.4_ai_assisted> & "C:\Program Files\Python313\python.exe" c:/Users/DELL/6.4_ai_a
sisted/task-5
Enter number of items to add: 4
Enter item name: Eggs
Enter item price: 130
Enter item quantity: 1
Added 1 x Eggs at 130.0 each.
Enter item name: Bread
Enter item price: 49
Enter item quantity: 2
Added 2 x Bread at 49.0 each.
Enter item name: Peanu Butter
Enter item price: 149
Enter item quantity: 3
Added 3 x Peanu Butter at 149.0 each.
Enter item name: Lays
Enter item price: 50
Enter item quantity: 4
Added 4 x Lays at 50.0 each.
Do you want to remove any item? (yes/no): no
10% discount applied.
Total bill after discount: 787.50
PS C:\Users\DELL\6.4_ai_assisted>
```